

Task 02 - python self learning

Link to your starting repo - <https://classroom.github.com/a/m0ZEczri>

Relevant files to read before starting

- exercise.md — The exercises you need to solve, naming conventions, grading, and AI usage policy.
- README.md — Environment setup, repository structure, and submission instructions.
- WORK-INSTRUCTIONS.md — Git workflow: how to create branches, open pull requests, and resolve conflicts.
- PRACTICES.md — Clean code guidelines and examples. Your code will be graded against these.
- GITHUB-CHECKS.md — Explanation of the automated Pylint and Pytest checks that run on every pull request.
- AI_RULES.md — Rules for using AI assistance. If you use AI, your first prompt must be the contents of this file.

AI Usage Policy

When you use an AI assistant for these exercises:

- Your **first prompt** must be the full contents of `AI\_RULES.md`.
- Upload all AI conversations **as PDF files** to a folder called `ai\_conversations/` in your repository. Any filename is acceptable. At least one conversation file must be submitted.

The Task

“Yam mesika” is a Hebrew python tutorial that contains exercises and explanations.

<https://github.com/PythonFreeCourse/Notebooks>

You need to hand in these 11 exercises from weeks 5,6,7 of the notebooks. If your python is rusty, I recommend reading week 3 and week 4 as well.

5.1 - אין לי וינגרט (no_vinnigrete), זו הדרך (thats_the_way)

5.2 - חתכת עוגה (piece_of_cake), cup of join(cup_of_join)

5.3 - לחששנית (parsle_tongue)

5.4 - כלים שלובים (interleaves, generator_interleaves)

Watch a video about [aspect oriented programing](#), and do these exercises -

6.2 - ריצת 2000 (running_2000)

6.3 - חתול ארוך הוא ארוך (long_cat_is_long)

6.4 - זכרו זכרו (remember_remember)

6.5 - group_by (group_by)

7.2 - צב שולח (sent_turtle)

Bonus exercises

Here are three bonus exercises about decorators -

1. Make a decorator factory which returns a decorator that decorates functions with one argument. The factory should take one argument, a type, and then returns a decorator that checks if the parameter the function receives is of the correct type. If it is wrong, it should raise a custom error. Read about `functools.wraps` and use it. In this exercise, online solutions will not be accepted. **The file name should be “type_check.py”, and the decorator factory should be called “type_check”.**

2. Write a decorator that will make the function print “surprise!” instead of its original functionality. **Pay attention - the file name should be “surprise.py” and the decorator name should be “surprise”**

3. Write a decorator that executes the function it wraps twice. Read about `decorator.decorator` and use it in this exercise. **Pay attention - the file name should be “twice.py” and the decorator name should be “twice”.**

How to hand in

- Hand in the assignments in the git repository you have created from the supplied link (Available in moodle). Use “How to open a pull request.pptx” provided slide if you are stuck.
- Use the file names in this format – <week>.<part>.<exercise name>.py
 - The exercise name is specified in brackets near the exercise name in hebrew
 - For example, for "זו הדרך", the file name would be - “5.1.thats_the_way.py”.
 - Make sure that the function name inside the file, is the in the name of of the exercise
 - “5.1.thats_the_way.py” should contain a function called “thats_the_way”
- There are some small exercises inside each section, you don’t have to do them or hand them in. Hand each exercise on a separate module unless the exercises are dependent on each other.
- make sure to use `__name__ == '__main__'` exists in every file
- Hand each exercise on a separate module unless the exercises are dependent on each other.
- **Pay attention, .ipynb file format will not be accepted, only .py files.**

Grading

- Each exercise grants 9 points, each bonus will give you up to 3 bonus points.
- In addition, you will be graded on your clean code. Look at the “clean code guidelines.docx” for additional information.

As usual, feel free to contact the course staff using slack.